

Influence of vegetation change on drip water dynamics and geochemical proxies in a shallow karst cave system

Yina Lyu^{a,b,c}, Weijun Luo^{b,c,*}, Reni Ustiatik^{d,e}, Guangjie Luo^a, Alena Gessert^f, Guangneng Zeng^g, Yanwei Wang^h, Jia Chen^{b,c}, Anyun Cheng^{b,c}, Shijie Wang^{b,c}

^a Guizhou Provincial Key Laboratory of Geographic State Monitoring of Watershed, School of Geography and Resources, Guizhou Education University, Guiyang 550018, China

^b State Key Laboratory of Environmental Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences, Guiyang 550081, China

^c Puding Karst Ecosystem Research Station, Chinese Academy of Sciences, Puding 562100, China

^d Soil Science Department, Faculty of Agriculture, Universitas Brawijaya, Jl. Veteran Malang 65145, East Java, Indonesia

^e International Research Centre for the Management of Degraded and Mining Lands, Universitas Brawijaya, Jl. Veteran Malang 65145, East Java, Indonesia

^f Institute of Geography, Faculty of Science, Pavol Jozef Šafárik University, Jesenná 504001 Košice, Slovakia

^g School of Eco-Environmental Engineering, Guizhou Minzu University, Guiyang 550025, China

^h School of Geography and Planning, Ningxia University, Yinchuan 750021, China

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ABSTRACT

Cave drip water is a fundamental carrier of climatic and environmental information, and its geochemical signatures are widely employed in speleothem-based paleoclimate reconstructions. While are well established, the role of rainfall intensity and vegetation-driven infiltration dynamics in shaping these signals remains insufficiently constrained. To address this, we carried out seven years of monitoring in Shawan Cave, a shallow karst cave system in Southwest China, integrating rainfall observations, continuous drip rate measurements at 30-minute resolution, and stable isotopes and elements/Ca data from drip water. Our results showed that drip water was present in both the dry and wet seasons from 2017 to 2019, whereas it was intermittently interrupted from 2020 to 2023 drip water, with more observed during the wet season and less during the dry season. This may be attributed to the deeper root systems of vegetation, which alter soil moisture and the amount of rainfall required to penetrate the soil and generate infiltration into the karst cave system, thereby changing the water reservoirs feed drip water over time. We propose that vegetation recovery, by influencing evaporation, can modulate drip water flow paths and thereby drive the enrichment of drip water $\delta^{18}\text{O}$. Intriguingly, although drip water $\delta^{18}\text{O}$ during 2017–2018 did not fully reflect changes in rainfall $\delta^{18}\text{O}$, during 2022–2023 it appeared to inherit the $\delta^{18}\text{O}$ signal of heavy rainfall. This indicates that drip water $\delta^{18}\text{O}$ has gradually evolved from reflecting annual or multi-year rainfall amounts to primarily recording the heavy rainfall of the summer monsoon. These observations demonstrate that drip water signals are not simply a direct record of rainfall but are mediated by subsurface hydrological processes. We propose that the combined effects of reservoir-feeding drip water and flow path change over time are critical in governing the transmission of rainfall signals into cave environments. Recognizing the influence of vegetation change and reservoir evolution refines the interpretation of speleothem proxies, enabling more reliable reconstructions of monsoon variability at sub-annual to interannual timescales.

1. Introduction

Speleothems have become one of the most important archives for paleoclimate reconstruction, prompting the scientific community to examine the hydroclimatic significance of multiple proxies, particularly $\delta^{18}\text{O}$, $\delta^{13}\text{C}$, and element/Ca ratios (Baker et al., 2019, Demény et al.,

2017, Duan et al., 2023). Modern cave monitoring of the $\delta^{18}\text{O}$, $\delta^{13}\text{C}$, and element/Ca in cave drip water and precipitation has revealed the mechanisms by which external climate information is transferred to speleothems (Baldini, 2010, Bernal et al., 2023, Duan et al., 2016, Lambert et al., 2010, Li et al., 2011, Matthey et al., 2010, Yin et al., 2021), particularly the links between seasonal rainfall, vegetation biomass, and

* Corresponding author.

E-mail address: luoweijun@vip.gyg.ac.cn (W. Luo).

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speleothem proxies (McNeill-Jewer et al., 2019; Railsback et al., 2019).

Previous studies have frequently used the relationship between rainfall $\delta^{18}\text{O}$ and speleothem $\delta^{18}\text{O}$ to reconstruct monsoon variability associated with orbital forcing (Cheng et al., 2016; Hu et al., 2008; Liu et al., 2020). In some caves, drip water $\delta^{18}\text{O}$ values remain relatively stable compared with rainfall $\delta^{18}\text{O}$, reflecting long residence times and signal integration (Li et al., 2011; Riechelmann et al., 2011; Genty et al., 2014). In others, drip water $\delta^{18}\text{O}$ tracks rainfall seasonality more closely (Cruz et al., 2005; Cobb et al., 2007; Pape et al., 2010) suggesting a more direct hydrological transfer. These contrasting behaviors indicate that speleothem $\delta^{18}\text{O}$ can preserve both smoothed and event-scale rainfall signals, depending on site-specific hydrology (Pérez-Mejías et al., 2018). Increasingly, research has expanded to include $\delta^{13}\text{C}_{\text{DIC}}$ and element/Ca ratios to evaluate the effects of rainfall variability and vegetation dynamics (Hess et al., 2024). For example, cave water $\delta^{13}\text{C}_{\text{DIC}}$ reflects wet/dry conditions resulting from rainfall-driven recharge, vegetation biomass, and soil CO_2 -production (Li et al., 2021; Yin et al., 2021; Wu et al., 2020). Similarly, the balance between rainfall amount and evapotranspiration exerts a dominant control on the variability in Mg/Ca and Sr/Ca ratios which display higher values during dry years and lower values during wet years (Cheng et al., 2022).

Several studies have emphasized the role of storm events in shaping drip water hydrochemistry through piston flow, dilution, and soil CO_2 exchange (Pu et al., 2014). During rainfall events, the dissolution of calcite consumes CO_2 from cave air, lowering cave air $p\text{CO}_2$, whereas during dry periods, unsaturated epikarst allows soil air CO_2 to enter the cave and raise cave air $p\text{CO}_2$ (Cao et al., 2021; Cuezva et al., 2011). These processes further influence drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element/Ca. To investigate the control of rainfall amount on drip water proxies, it is necessary to determine how drip water responds to rainfall amount. The log-linear correlation between the coefficient of variation and maximum discharge is used to classify the seepage flow modes in drip water (Baldini et al., 2006). However, rainfall amount must overcome a combination of evapotranspiration losses and subsurface moisture deficits, which influence drip discharge variability and thus control drip water proxies (Baker et al., 2021). This implies that vegetation biomass above caves can strongly influence drip discharge and geochemical variability, though the underlying mechanisms remain unclear.

Recent cave monitoring has advanced understanding of drip water behavior by combining multiple proxies ($\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element/Ca) to identify seasonal and annual patterns over multi-year timescales (Oster et al., 2012; Matthey et al., 2010; Yin et al., 2023; Bao et al., 2023). These studies typically emphasize oxygen and carbon isotopes to infer infiltration processes and vadose-zone residence times. However, comparisons of drip rates with rainfall volumes show that drip discharge often responds more strongly several days after rainfall events, and that this lag is not always equivalent to vadose-zone residence time (Ruiz et al., 2024). Drip water responds differently to rainfall inputs, indicating the existence of complexities not previously considered (Zhang and Li, 2019). Such complexities highlight the need for high-frequency drip rate and geochemical data across a range of rainfall intensities to test the sensitivity of $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element/Ca, and to improve the reliability of speleothem records in reconstructing short-term climate variability.

Despite these advances, important gaps remain. First, most studies have not fully addressed the role of flow path and reservoir feeding drip water in shaping the transfer of rainfall signals to drip water. Second, although vegetation change is recognized as a key control, few studies have directly linked vegetation recovery to changes in soil moisture thresholds and preferential flow that alter drip water dynamics. Third, high-resolution, long-term datasets that combine isotopes, elemental ratios, and continuous drip rate monitoring remain rare, limiting our ability to disentangle the joint effects of rainfall intensity and vegetation on drip water proxies.

Prior studies in Shawan Cave during 2017–2019 indicated that the initial stages of vegetation restoration impacted drip water $\delta^{18}\text{O}$,

$\delta^{13}\text{C}_{\text{DIC}}$, and ion concentrations (Lyu et al., 2020). In contrast, this study explores the effects of vegetation transition (from low shrub to approximate mature shrub or tree) over longer timescales on drip water dynamics and their associated geochemical indicators. Here we present a seven-year study of rainfall and vegetation influences on drip rates and drip water geochemistry in Shawan Cave, Guizhou Province, Southwest China. This shallow karst system is overlain by vegetation that has naturally recovered from cultivated crops (corn and canola) to dense shrubs (Lyu et al., 2020). Our dataset includes drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca measurements, with weekly sampling during 2018 and 2022 and drip rate data recorded every 30 min during rainfall events. The main objective is to evaluate how rainfall intensity and vegetation change regulate drip water discharge and proxy variability. This study provides critical baseline information for interpreting speleothem records of monsoon variability, an area where long-term, high-resolution monitoring remains limited.

2. Site description

2.1. Climatic setting

The Shawan Cave (26.36°N, 105.76°E), located in Southwest China (Fig. 1a), is characterized by a subtropical monsoon climate with humid summers and dry winters. Historical data from the China Meteorological Data Network show that over 30-year period from 1981 to 2010, the annual rainfall above Shawan Cave was 1,341 mm, with a mean annual air temperature of 15.2 °C. During the monitoring period from 2017 to 2021, the annual mean surface air temperature was 17.4, 16.4, 16.3, 16.3, and 16.4 °C, respectively, and the annual rainfall amounts were 1,166, 1,408, 1,064, 1,403, and 1,013 mm, respectively (Lyu et al., 2023). Summer rainfall (April–October) accounts for approximately 80% of the annual rainfall, which is associated with large-scale circulation, including the summer monsoon circulation and distant moisture sources (Indian Ocean). In contrast, winter rainfall is lower and is linked to local evaporation and nearby moisture sources (West Pacific and South China Sea) (Yin et al., 2023; Zhang et al., 2019). The shift in vegetation overlying Shawan Cave from crops to natural vegetation in 2017 has resulted in the current composition, which is primarily trees, shrubs, and grasslands (Wang et al., 2019).

2.2. Cave setting

Shawan Cave developed in the limestone of the Middle Triassic (T₂ g) Guanling Formation, situated at a depth of 6–8 m below the limestone surface, with a soil cover exceeding 50 cm (Fig. 1b). The cave has a single entrance, approximately 30 m long, 3 m wide, and 3 m high, with the entrance located at 1,170 m above sea level. Five drip water monitoring sites (1#, 2#, 3#, 4#, and 5#) were selected for observation within the entrance, gallery, and chamber of Shawan Cave, where stalactites, stalagmites, and flowstones are present (Fig. 1c). These drip sites are seasonal drips, that respond to rainfall. Microenvironmental monitoring in Shawan Cave was performed. During the monitoring period (2017–2019), the annual mean cave air temperature was 17.4 °C, and cave relative humidity remained consistently at 100% (Lyu et al., 2020).

3. Methods

3.1. Meteorology, vegetation, and cave microenvironments monitoring

During the monitoring period, spanning from 2017 to 2023, we used the meteorological station located above Shawan Cave to record surface air temperature and rainfall amount every hour. Vegetation biomass (W) was calculated using the allometric equation $W = a(D^2H)^b$ (Liu et al., 2009), where D is the breast-height diameter, H is the height, and a and b are species-specific constants. To obtain D, we analyzed the growth

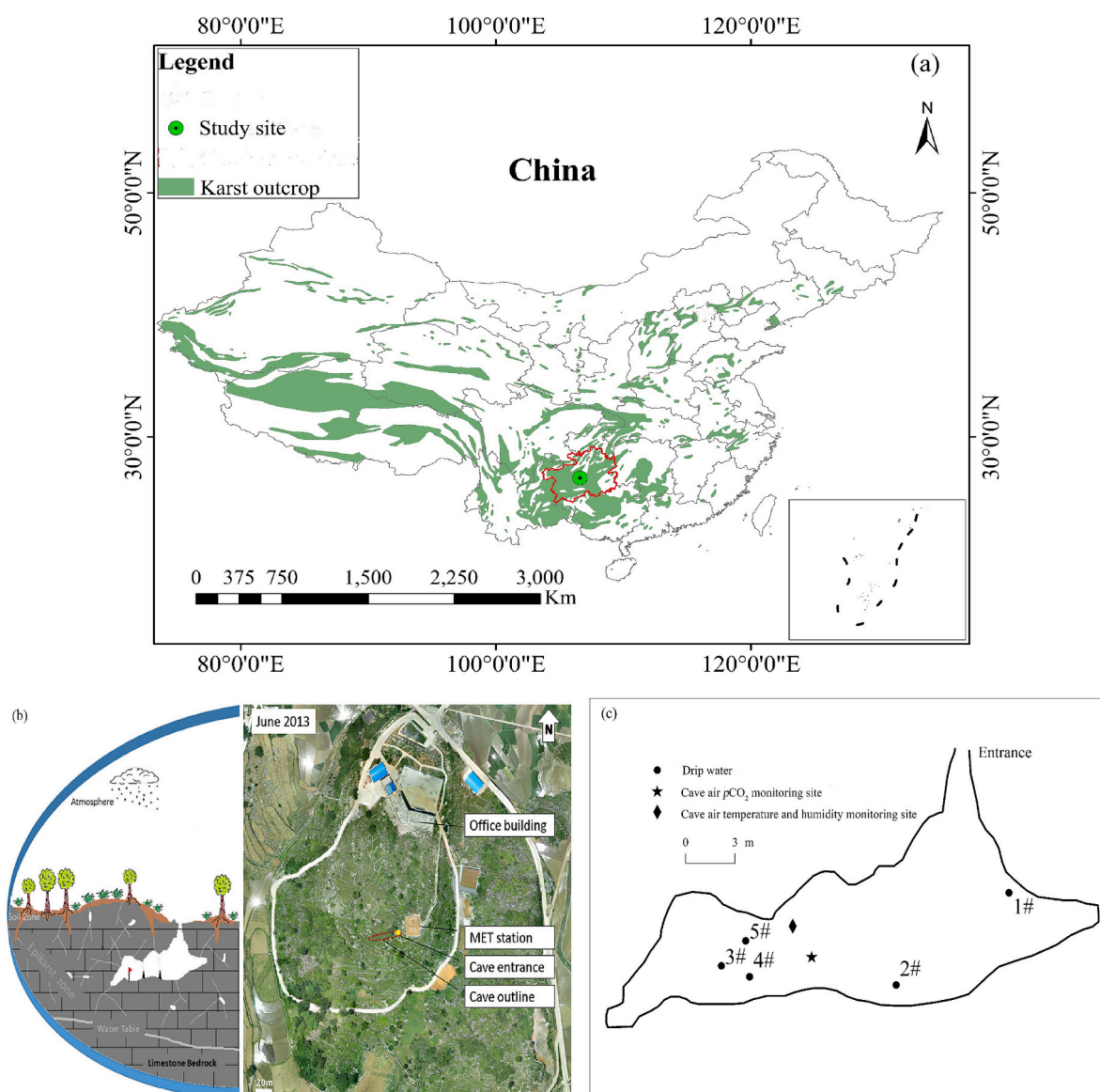


Fig. 1. (a) Locations of the study area (red circle) and Shawan Cave (small green circle). (b) Environmental monitoring conditions outside Shawan Cave. (c) Monitoring sites inside Shawan Cave. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

rings of one hundred trees in the Shawan plot (outside Shawan Cave) for the period 2018–2022. These trees belonged to 19 species. A flux tower equipped with an eddy covariance system was installed in the plot to measure net ecosystem exchange (NEE) and evapotranspiration. Gross primary production (GPP) was then estimated from the NEE data. The monthly effective rainfall was calculated as monthly rainfall minus monthly evapotranspiration. The soil moisture at a depth of -10 cm in the soil overlying Shawan Cave was monitored using a multiplexed automated soil gas flux system (LI-8150, LiCor, USA). The soil moisture measurements were accurate within $\pm 1\%$.

We employed a CCIA-DLT-EP CO₂ carbon isotope analyzer (Los Gatos Research, USA) to measure cave air pCO₂ continuously, with a measurement range of 380–25,000 ppm and a frequency of 1 Hz. Cave air pCO₂ measurements were taken from 2017 to 2021. Drip rates presented in this study were recorded during two periods: from June 2017 to December 2018 and from June 2022 to December 2023. Since drip water flowed continuously during 2018–2022, yet was intermittently interrupted during 2022–2023, we compared the patterns of drip rate changes in the two periods. We used automatic drip loggers (Stalagmite MK3) to record drip rates at five drip sites, with a half-hour resolution.

The detection limit of the Stalagmite MK3 drip logger is 9000 drips/30 min.

3.2. Rainwater and drip water sampling

Rainwater samples were collected using standard gauges in the rainy seasons (April–October) during 2017–2023. These samples were filtered through 0.45 μ m Millipore filters and transferred into brown glass bottles after rainfall events. Drip water samples were collected weekly between June and August 2017 and between June and August 2022. During the rest of the monitoring period, drip water samples were collected monthly. Drip water samples were collected at each site, with containers left in place for 0.5–1 h during the rainy season and 8 h during the dry season. The rainwater and drip water samples were poured into acid-cleaned brown glass bottles, stored in a refrigerator (approximately 4 °C) before analysis. The rainwater samples in bottles of 30 ml were analyzed for $\delta^{18}\text{O}$. The drip water samples were in bottles of different sizes, which were divided into three groups, with 30 ml allocated for the analysis of $\delta^{18}\text{O}$, 50 ml for the analysis of $\delta^{13}\text{C}_{\text{DIC}}$, and 30 ml for the elemental analyses, respectively. For the elemental analysis

subsample, ultrapure HNO_3 (1:1) was added to acidify the sample to $\text{pH} < 2$. Drip water DIC were determined by immediate on-site titration in the cave, using an Alkalinity test (Merck KGaA Company, Germany) (accuracy: 0.1 mmol/L).

3.3. Analysis of $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and elements concentrations in water samples

Rainwater and drip water $\delta^{18}\text{O}$ were measured using a DLT-100 liquid water isotope analyzer (IWA-35EP; Los Gatos Research, USA), with an accuracy of $\pm 0.1\text{‰}$ for $\delta^{18}\text{O}$. The oxygen isotope data are reported relative to Vienna Standard Mean Ocean Water. Drip water $\delta^{13}\text{C}_{\text{DIC}}$ was measured using a MAT-252 mass spectrometer, with an analytical uncertainty of less than $\pm 0.10\text{‰}$ for $\delta^{13}\text{C}$ values. The measured values are reported relative to Vienna Pee Dee Belemnite. Concentrations of Ca^{2+} , Mg^{2+} , and Sr^{2+} were analyzed using inductively coupled plasma optical emission spectroscopy (ICP -OES; Vista MPX

type), which is Vista MPX type. The detection limits for Ca^{2+} , Mg^{2+} , and Sr^{2+} are 0.003, 0.005, and 0.008 ppm, respectively, and concentrations were expressed in mg/L.

4. Results

4.1. Rainfall, vegetation, soil moisture, rainfall $\delta^{18}\text{O}$, and cave air pCO_2

The mean annual rainfall outside Shawan Cave was 1,182 mm from 2017 to 2023. The annual rainfall in 2018 and 2020 was 1,408 and 1,403 mm, respectively, which were 226 and 221 mm higher than the 7-year mean. The annual rainfall in 2019, 2021, and 2022 were 1,064, 1,013 mm, and 1071 mm, respectively, which were 118, 169, and 111 mm lower than the 7-year mean value. Rainfall in 2017 and 2023 was 1,166 and 1,150 mm, which is close to the 7-year mean (Fig. 2h). The biomass overlying Shawan Cave increased significantly, from 20.83 tons per hectare in 2017 to 41.07 in 2022 and 60.66 in 2024. Both the GPP in

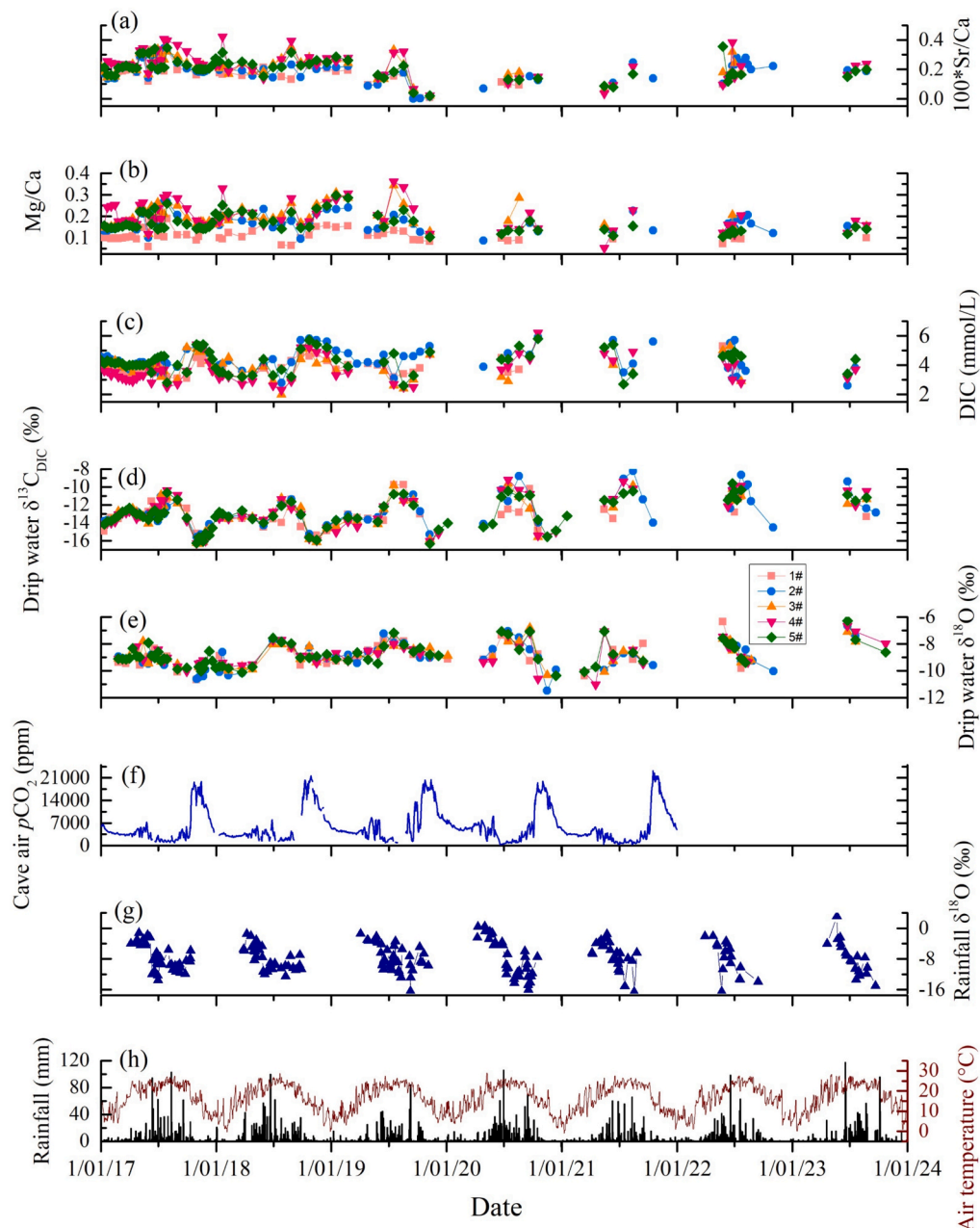


Fig. 2. Time series of drip water Sr/Ca (a), Mg/Ca (b), $\delta^{13}\text{C}_{\text{DIC}}$ (c), $\delta^{18}\text{O}$ (d), cave air pCO_2 (e), rainfall $\delta^{18}\text{O}$ (f), and rainfall amount (g) during 2017–2023.

the Shawan plot (2016–2023) and the breast-height diameters of its trees (2018–2023) showed an increasing trend (Fig. 3a and b). These trends suggest that the vegetation outside Shawan Cave has improved and is vegetation restoration situation. With increased vegetation cover, effective rainfall was relatively high from 2017 to 2020 but relatively low from 2021 to 2023; soil moisture followed this same pattern, increasing from 2018 to 2020 but has decreased since 2021 (Fig. 3c and d).

Although data on rainfall $\delta^{18}\text{O}$ were absent in the dry season, rainfall $\delta^{18}\text{O}$ showed significant seasonality, with negative values in summer and positive values in autumn. The mean rainfall $\delta^{18}\text{O}$ values for the summer months (April–October) in 2018 and 2020 were -7.98‰ and -7.83‰ , respectively, showing a more pronounced negative trend. The mean summer rainfall $\delta^{18}\text{O}$ value in 2021 was -6.96‰ , which was more positive. The mean summer rainfall $\delta^{18}\text{O}$ values in 2017, 2019, 2022, and 2023 were -7.73‰ , -7.76‰ , -7.77‰ , and -7.66‰ , respectively, and remained stable (Fig. 2g). Annual variations in rainfall $\delta^{18}\text{O}$ were consistent with rainfall amount. Higher rainfall corresponded to more negative rainfall $\delta^{18}\text{O}$, while lower rainfall was associated with more positive $\delta^{18}\text{O}$ values. The rainfall $\delta^{18}\text{O}$ remained stable in years when annual rainfall amount was relatively close to the mean (Fig. 2g and h).

The cave air $p\text{CO}_2$ in Shawan Cave exhibited significant seasonal variation, with lower values in the spring and summer and higher values in autumn and winter. Cave air $p\text{CO}_2$ reached its highest values from mid-October to early December. The mean values from 2017 to 2021 were 15,505, 15,653, 16,414, 15,693, and 16,464 ppm, respectively, showing an overall increasing trend. Cave air $p\text{CO}_2$ reached its lowest values during June–August. The mean values from 2017 to 2021 were 1758, 2911, 2529, 1577, and 1201 ppm, respectively, showing higher

values during 2017–2019 and lower values during 2020–2021 (Fig. 2f). The cave air humidity in Shawan Cave remained constant 100% throughout the study period.

4.2. Drip rate and drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, DIC, and elements/Ca during 2017–2018

During 2017–2018, when rainfall was absent from December to March, the drip rates at 1#, 2#, 3#, 4#, and 5# were close to zero or above, suggesting that the water reservoir in the vadose zone may have lost a small amount of infiltration water. From April to November, as rainfall gradually decreased, the variability in drip rate at 4# and 5# closely resembled the rainfall pattern. When rainfall decreased, the drip rate at 1# and 3# approached zero. Conversely, when rainfall increased, the drip rate at 1# and 3# rose rapidly. During May–August 2018, the mean drip rates at 1#, 2#, 3#, 4#, and 5# were 421.2, 266.9, 411.3, 923.6, and 2537.3 drips/30 min, respectively. Notably, the observed drip rate variability at site 2# was distinct from the others, showing a relatively slower but continuous response to rainfall (Fig. 4a). This suggests that site 2# consistently responds to cumulative rainfall.

The sampling frequency was once a week from June 2017 to February 2018 and once a month from March to December 2018. We collected sufficient drip water for the measurement of $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element concentrations. Drip water $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca values decreased from August 2017 to December 2017, which was related to an increase in cave air $p\text{CO}_2$. Although drip water $\delta^{18}\text{O}$ decreased during the period, its variability was more closely associated with rainfall $\delta^{18}\text{O}$. Time series of drip water $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca data are presented in Fig. 2. $\delta^{18}\text{O}$ and $\delta^{13}\text{C}_{\text{DIC}}$ values were higher in summer and lower in

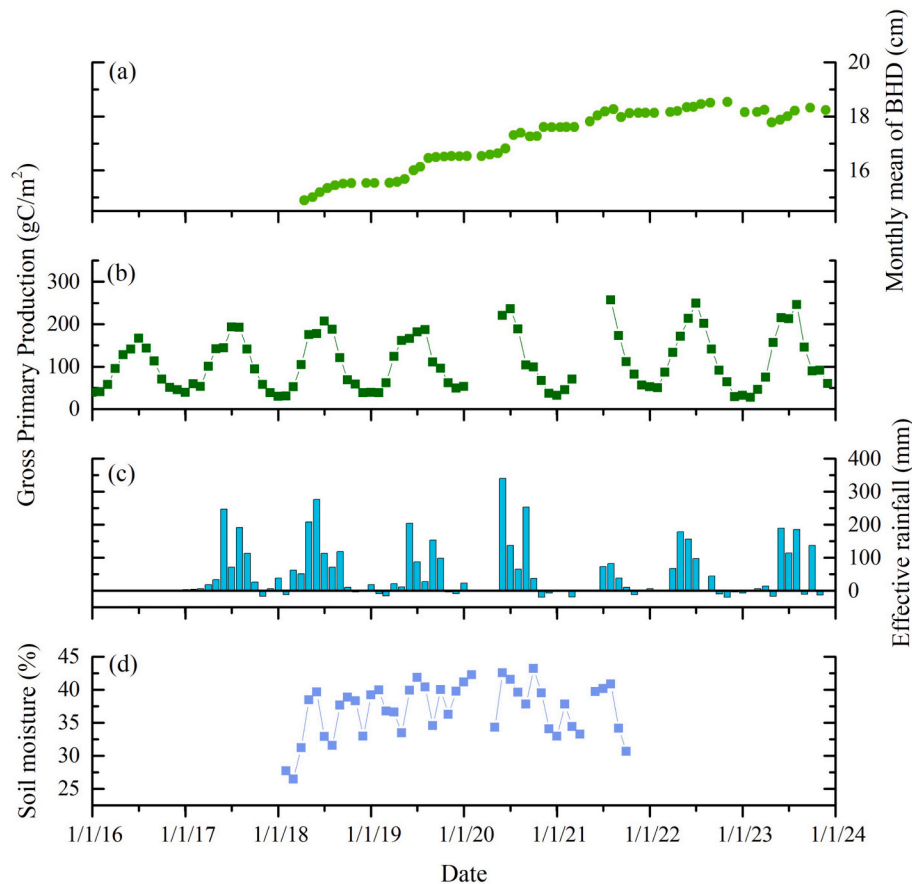


Fig. 3. Time series of monthly mean of breast-height diameter (BHD) for vegetation in Shawan study area during 2018–2023 (a). Time series of Gross Primary Production (GPP) for vegetation in Shawan study area during 2016–2023 (b). Time series of monthly effective rainfall in Shawan study area during 2017–2023 (c). Time series of monthly soil moisture in Shawan study area during 2018–2021 (d).

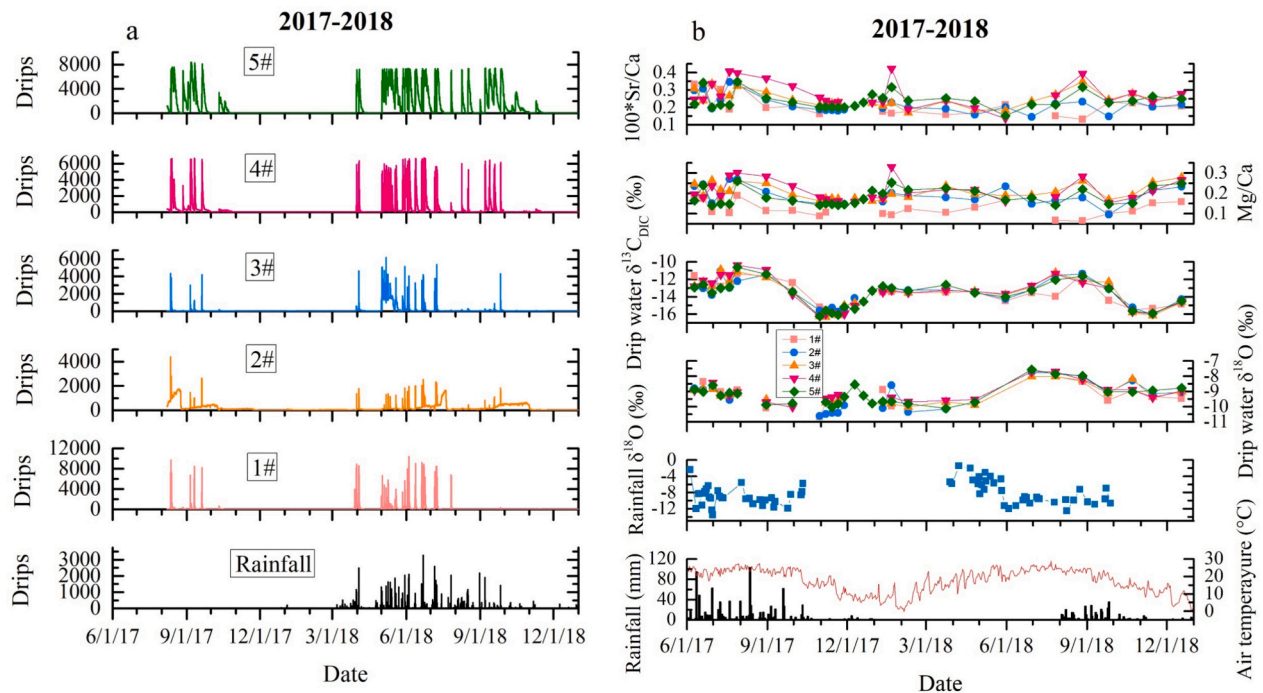


Fig. 4. (a) Variations in rainfall drips and drip rates at sites 1#, 2#, 3#, 4#, and 5# during 2017–2018. Drips from rainfall represent only the intensity of rainfall events. (b) Variations in rainfall and rainfall $\delta^{18}\text{O}$, alongside variations in drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca during 2017–2018.

winter, consistent with cave air $p\text{CO}_2$ (Fig. 4b). This suggests that high-frequency sampling of drip water can capture more specific climate signals. Drip water DIC showed an increasing trend during 2017–2023, reflecting vegetation restoration (Fig. 2c).

4.3. Drip rate and drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and elements/Ca during 2022–2023

A rise in drip water rate was observed when rainfall exceeded 20 mm between June 2022 and August 2022. From September 2022 to May

2023, when rainfall was below 20 mm, the drip rate at 1#, 2#, 3#, 4#, and 5# were consistently zero. These observations indicate that some infiltration water in the reservoir may be lost, resulting in less water moving downward into the cave to form drip water. During June–December 2023, there were three periods with rainfall below 20 mm: July–August, September–October, and November–December. Consequently, the drip rates at 2# and 3# were close to zero during all three periods, while those at 4# and 5# were close to zero during September–October and November–December. The drip rates at 2#, 3#, 4#, and 5# persisted during the rest of the period. During May–August 2023,

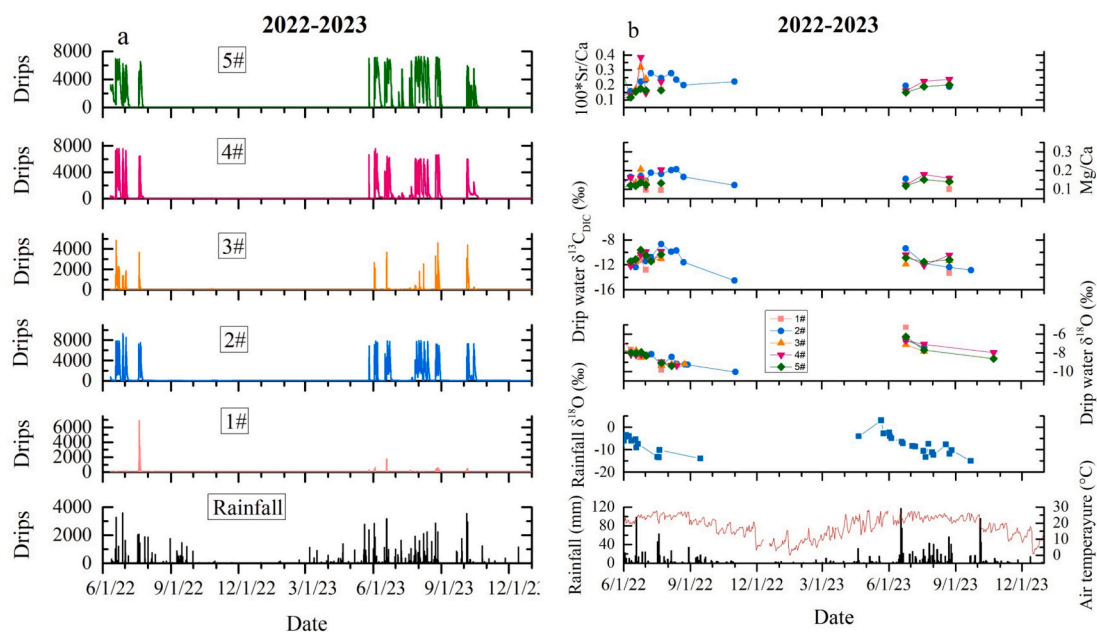


Fig. 5. (a) Variations in rainfall and drip rates at sites 1#, 2#, 3#, 4#, and 5# during 2022–2023. Drips from rainfall represent only the intensity of rainfall events. (b) Variations in rainfall and rainfall $\delta^{18}\text{O}$, alongside variations in drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca during 2022–2023.

the mean drip rates at 1#, 2#, 3#, 4#, and 5# were 9.9, 1,076.5, 76.2, 1,105.9, and 1,948.9 drips/30 min, respectively. Notably, the drip rate at 1# was close to zero during July–December 2018, except for four peaks exceeding 2,000 drips (Fig. 5a).

The sampling frequency was once a week from June to September 2022, and the sampling frequency was once a month from July to December 2023. Compared with drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca values during 2017–2018, the variations in drip water proxies during June–October 2022 differed from those during June–October 2017. During June–October 2022, drip water $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca values all exhibited increasing followed by decreasing trends, suggesting that cave air $p\text{CO}_2$ likely influenced these proxies. However, drip water $\delta^{18}\text{O}$ showed a decreasing trend during the period, consistent with rainfall $\delta^{18}\text{O}$ variability. During December 2022–June 2023, we did not collect drip water for $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element concentrations (Fig. 5b). In other words, drip water was interrupted during the dry season, implying that infiltration water in the vadose zone cannot generate recharge during this period. We found that drip water interruption commenced in February 2020 based on the time series of $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca data. The timing of drip water interruption advanced year by year, as illustrated in Fig. 2.

4.4. Drip rate variability at different rainfall intensities

A generalized additive model (GAM) was used to analyze the rainfall threshold influencing drip rates (Wood, 2001). The model determined thresholds of 9.43, 7.24, 9.04, 9.5, and 9.35 mm for the five drip sites, respectively. Notably, when rainfall exceeds 40 mm, the drip rate of five sites exhibits significant variation (Fig. S1 in Supplementary Material). Therefore, rainfall categories are defined based on the generalized additive model: Light rainfall: three consecutive days with daily rainfall below 10 mm. Moderate rainfall: five consecutive days with daily rainfall between 10 mm and 40 mm. Heavy rainfall: five consecutive days with daily rainfall exceeding 40 mm.

During light rainfall, when the rainfall amount for three consecutive days in 2018 was 0.6 mm. The drip rates at 2# and 3# remained

constant, with mean values of 102.1 and 25.2 drips/30 min, respectively. The drip rates at 4# and 5# decreased, with mean values of 4.2 and 24.6 drips/30 min, respectively. The drip rate at 1# was consistent with rainfall (Fig. 6a). This suggests that less infiltration water was required to saturate the soil and subsequently feed drip sites 2#, 3#, 4#, and 5# during light rainfall in 2018. When the rainfall over five consecutive days in 2022 totaled 1 mm. The mean drip rate at 2# was 35.3 drips/30 min, whereas the drip rates at 2#, 3#, 4#, and 5# were all close to zero. This suggests that during light rainfall in 2022, infiltration water recharged only drip site 2# (Fig. 6b).

During moderate rainfall, on three consecutive days in 2018 when the rainfall amount was 31.1 mm. The drip rates at sites 1#, 2#, 3#, 4#, and 5# sites. The mean drip rates at these sites were 8.8, 46.9, 24.2, 3.8, and 11.6 drips/30 min, respectively (Fig. 7a). This shows that the reservoirs feeding the five drip sites can recharge at 2# drip sites may be the largest. During moderate rainfall in 2022, the amount over three consecutive days was 31.4 mm. The drip rate at 1# was nearly zero and did not respond until rainfall reached 7.2 mm. The drip rates at other sites remained stable, with mean values of 88.9, 2.3, 7.2, and 3.0 drips/30 min for 2#, 3#, 4#, and 5#, respectively (Fig. 7b). Despite moderate rainfall, infiltration water fed only drip site 2#, suggesting a different recharge pathway at this site. This pathway is potentially attributed to bypass flow or smaller water storage.

Rainfall over three consecutive days in 2018 reached 48.2 mm, which was regarded as heavy rainfall. The variation patterns in drip rates at 1#, 2#, and 3# were analogous to rainfall, exhibiting two peaks and remaining stable in the absence of rainfall. In addition, the drip rate at 4# showed two peaks with fluctuating changes. However, the drip rate at 5# decreased during rainfall and increased in its absence, opposite to the patterns at other sites. The mean drip rate at 1#, 2#, 3#, 4#, and 5# were 1,885.1, 599.3, 409.5, 1,794.0, 5,724.2 drips/30 min, respectively (Fig. 8a). This suggests the existence of a significant karst water store with stable and higher flow capacity during 2018. During heavy rainfall in 2022, when the total rainfall over three consecutive days was 79.4 mm, the drip rate variabilities at sites 1# and 3# in last three days were in phase with rainfall, showing higher rates during

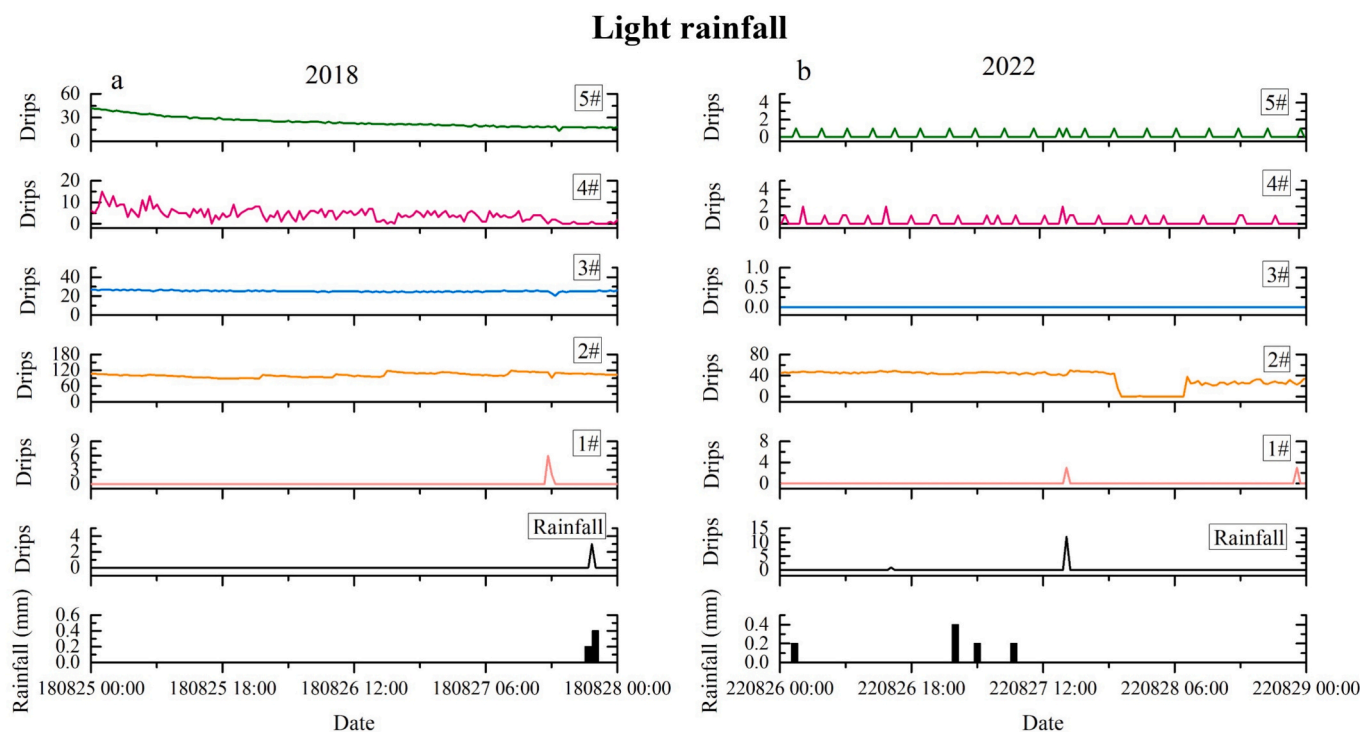


Fig. 6. Comparison of variations in rainfall and drip rates at sites 1#, 2#, 3#, 4#, and 5# during light rainfall in 2018 (a) and 2022 (b). Drips from rainfall represent only the intensity of rainfall events.

Moderate rainfall

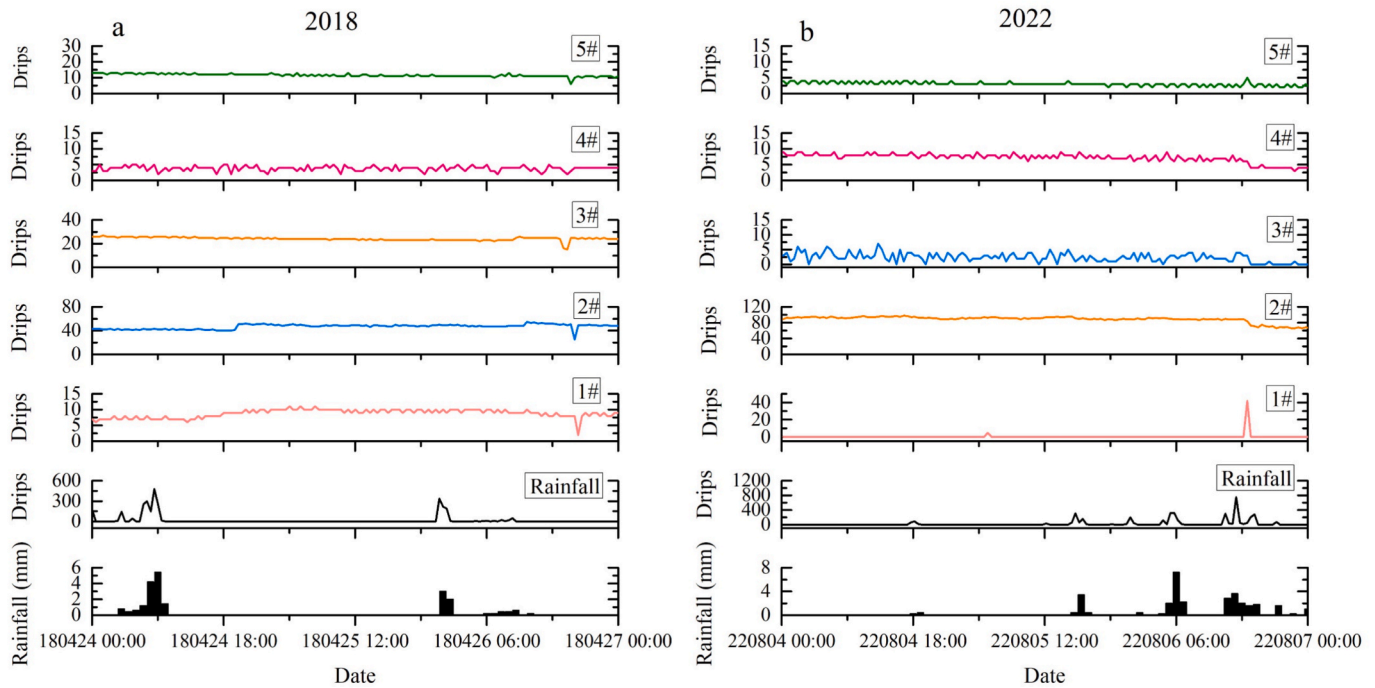


Fig. 7. Comparison of variations in rainfall and drip rates at sites 1#, 2#, 3#, 4#, and 5# during moderate rainfall in 2018 (a) and 2022 (b). Drips from rainfall represent only the intensity of rainfall events.

Heavy rainfall

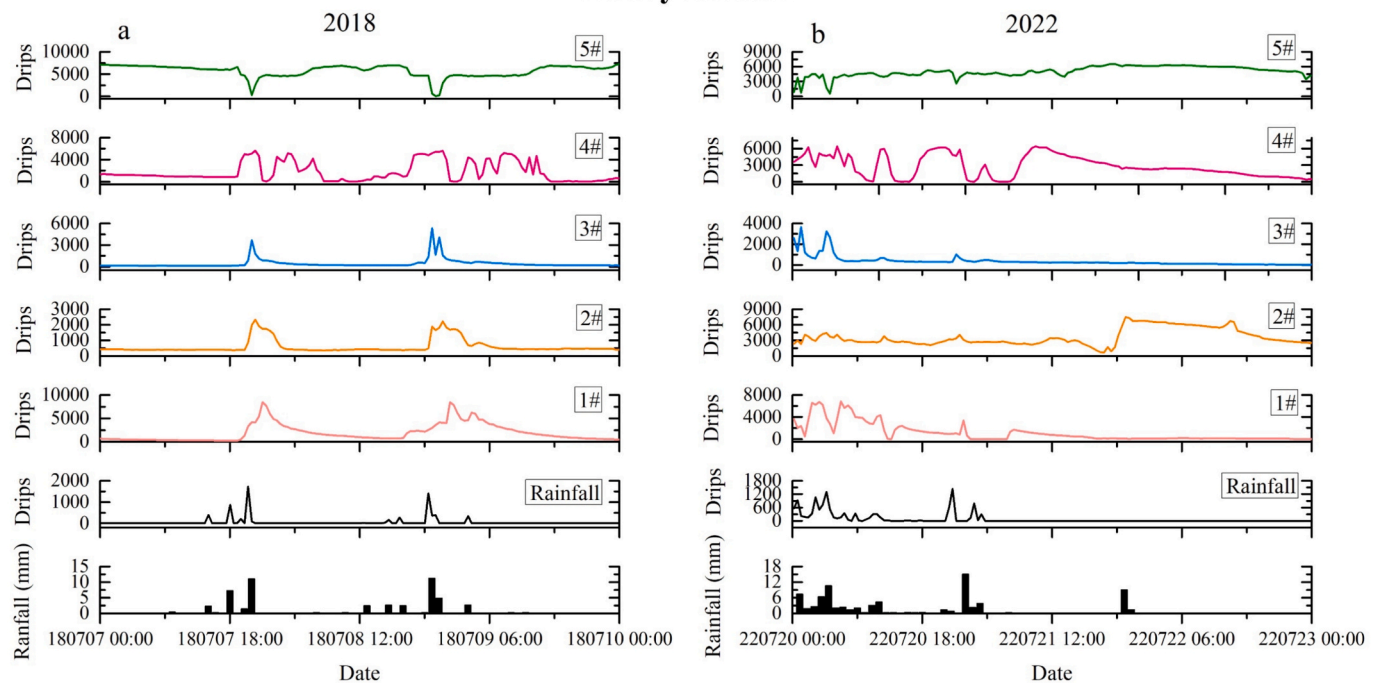


Fig. 8. Comparison of variations in rainfall and drip rates at sites 1#, 2#, 3#, 4#, and 5# during heavy rainfall in 2018 (a) and 2022 (b). Drips from rainfall represent only the intensity of rainfall events.

rainfall and zero rates in its absence. The drip rates at sites 2# and 5# remained stable during rainfall and increased in its absence. The drip rate variability at 4# oscillated during rainfall and decreased in its absence. The mean drip rates at 1#, 2#, 3#, 4#, and 5# were 1,103.9, 3,619.4, 372.0, 2,782.0, 5,002.6 drips/30 min, respectively (Fig. 8b).

These observations suggest the absence of a significant karst water reservoir and a lower flow capacity during periods without rainfall in 2022.

5. Discussion

5.1. Evolution of drip water infiltration and transmission over time

In karst systems, the primary factors influencing the drip rate are seasonal rainfall distribution, air temperature, water movement, and water reservoirs. Generally, drip water may be fed by different water movements, for example, rapid preferential flows (fissure and conduit), and slow matrix flow, or a combination of these. The speed of the drip rate depends on the relative proportions of matrix and preferential flows (Bradley et al., 2010). The reservoirs are formed by an upper soil layer (with vegetation cover) cover and a lower mixing layer, with reservoir size determining the drip water flux (Lases-Hernandez et al., 2019). In Shawan Cave, the patterns in the time series of breast-height diameter GPP, effective rainfall, and soil moisture could explain the drip water flux. During 2017–2020, sparser vegetation cover induced weak evapotranspiration, leading to higher effective rainfall and soil moisture, which in turn facilitated more drip water recharge to the cave. In contrast, during 2021–2023, increased vegetation cover led to strong evapotranspiration, resulting in lower effective rainfall and soil moisture, ultimately leading to the cessation of drip water.

Between the rainy seasons of 2018 and 2023, the drip rate at 1# and 3# decreased from 421.2 to 9.9 drips/30 min and from 411.3 to 76.2 drips/30 min, respectively, whereas the drip rate of 2# increased from 266.9 to 1,076.5 drips/30 min, and the changes at sites 4# and 5# were relatively small. Most studies have shown that the drip rate variability is primarily controlled by rainfall change (Duan et al., 2016; Pérez-Mejías et al., 2018). The relatively lower air temperature and similar rainfall (except for the lower rainfall in 2020–2021 and 2022–2023 dry seasons) observed outside Shawan Cave during the dry seasons of 2017–2023 (Fig. S2 in Supplementary Material) rule out the coupled rainfall-temperature relationship as the primary control on drip rates at this site. Hence, shifts in water movement and reservoirs may be the main factors influencing drip rate variability at these sites. The infiltration water in reservoirs feeding individual drip site has specific reservoirs or movement pathways (1#, 2#, 3#, 4#, and 5#), because the drip rate variations of the five sites over time are different (Baker et al., 2000). More detailed monitoring is required to explain the evolution of water movement and reservoirs over time controlling the recharge of different rainfall intensities to drip water in Shawan Cave.

The results under light rainfall in 2018 show that, except for site 1#, where the drip rate changes in phase with rainfall and may be fed solely by preferential flow, the drip rates at 2#, 3#, and 5# remained stable, implying that reservoirs feed the drip water. However, the drip rate variability at site 4# oscillated, suggesting it may be fed by both matrix and preferential flows and lacks a stable reservoir (Williams, 2008). By 2022, the drip rates at sites 1#, 3#, 4#, and 5# were close to zero, which could be attributed to deeper vegetation roots resulting from the gradual flourishing of vegetation between 2018 and 2022 (Lyu et al., 2020). The deeper vegetation roots may absorb water from the reservoirs or change seepage, influencing reservoir feeding of drip water (Breecker et al., 2012; Meyer et al., 2014). In contrast, the drip rate at site 2# remained stable, likely receiving water from multiple storage reservoirs, as bypass flow could be activated over time (Baker et al., 2021). Therefore, it is likely that the individual reservoirs feeding drip site 2# will increase over time.

During moderate rainfall in 2018, drip rates at sites 2#, 3#, and 5# remained stable, indicating that they are fed solely by reservoirs. The oscillations of drip rates at 1# and 4# further suggest that the two sites may not have reservoirs; however, they may be fed by preferential flow and by a combination of matrix and preferential flows, respectively. In contrast, the drip rates at sites 3#, 4#, and 5# demonstrated fluctuating variations in the moderate rainfall of 2022. The drip rate at 2# was high and stable, whereas the drip rate at 1# was nearly zero in the absence of rainfall and relatively higher in its presence. These results indicate that the 3# and 5# sites may gradually evolve into reservoirs that are more

sensitive to large rainfall due to deeper root systems driving preferential flow. The 2# sites may evolve into a reservoir consisting of multiple individual reservoirs, whereas drip water at sites 1# and 4# continues to be fed by preferential flow and by a combination of matrix and preferential flows, respectively.

Similar patterns of drip rate from 1#, 2#, 3#, and 4# with rainfall were observed in 2018 where the rainfall was heavy. Compared with the drip rate variabilities in 1#, 2#, and 3#, the drip rate at site 4# exhibits fluctuating change, which is attributed to different seepage pathways. As previously discussed, site 4# is fed by a combination of matrix and preferential flows, whereas sites 1#, 2#, and 3# are fed by preferential flow. Additional high-frequency and relatively stable changes in the drip rate at site 5# were inversely correlated with rainfall, indicating that drip water may be generated from overflow through a linked water reservoir as karst reservoirs fill and overflow. In contrast, during heavy rainfall in 2022, the patterns in drip rate at 1#, 3#, and 4# were consistent with those observed in 2018. However, a different drip rate pattern at site 2# was observed: its drip rate remained stable during rainfall and increased in its absence, showing in-phase variation with site 5#. These results demonstrate that the water movements at sites 1#, 2#, 3#, and 5# were preferential flows and the permeability at sites 1#, 3#, and 5# increased over time, owing to deeper root systems activating bypass preferential flows. The multiple individual reservoirs feeding site 2# are illustrated, showing possible pathways by which water might be delivered to the cave through preferential flow.

In summary, in 2018, the water movements at sites 2# and 3# in Shawan Cave belonged to preferential flows, the drip at site 5# may have been overflow-driven, and the 1# drip site exhibited matrix flow. Reservoirs fed sites 2#, 3#, and 5#, whereas site 1# had no reservoir, and site 4# was fed by a combination of matrix and preferential flows, potentially lacking a stable reservoir (Fig. 9a). By 2022, the reservoir feeding the 2# site had evolved from a single reservoir to multiple reservoirs (Fig. 9b). With the development of overlying vegetation, deeper root systems may activate bypass preferential flows due to the relatively thin cave roof. It was found that rainfall amount is not a main factor controlling seasonal interruption of drip water in Shawan Cave, where the ceiling is thin. However, these dynamic evolutions in water pathways, and their reservoir variability over time (Baker et al., 2000), result in seasonal interruption of drip water in Shawan Cave, with more drip water in the wet season and less or none in the dry season. These observations have important implications for drip water proxies, influencing $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element/Ca values.

5.2. Initial phase, later phase, and long-term variations of drip water proxies

Through previous observational research, it has been discovered that the ventilation type of Shawan Cave differs from the ventilation types of other caves in the same region, leading to a seasonal variation in cave air $p\text{CO}_2$ that is exactly opposite to that of soil air $p\text{CO}_2$. It is proposed that cave ventilation is the main factor controlling the seasonal variation in $\delta^{13}\text{C}_{\text{DIC}}$ values of drip water. Meanwhile, monitoring of $\delta^{18}\text{O}$ in rainfall and drip water suggests that hydrological processes linked to overlying vegetation changes may be another key driver of interannual $\delta^{18}\text{O}$ variability, alongside the rainfall signal itself. Given that the seasonal variation in cave air $p\text{CO}_2$ in Shawan Cave is opposite to the seasonal variation in rainfall, the seasonal changes in Mg/Ca and Sr/Ca ratios in the drip water are jointly influenced by rainfall and cave air $p\text{CO}_2$ (Lyu et al. 2020, 2023).

However, drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element/Ca values are likely to vary according to the residence time of water in the vadose zone, which is controlled by different rainfall intensities (Robertson and Gaziz, 2006). This may result in different isotopic inputs to individual components of the karst water flow systems, and different degrees of water–rock reactions affecting the input of isotopes and elements from bedrock (Drăguşin et al., 2017; Hu et al., 2021; Oster et al., 2021). These

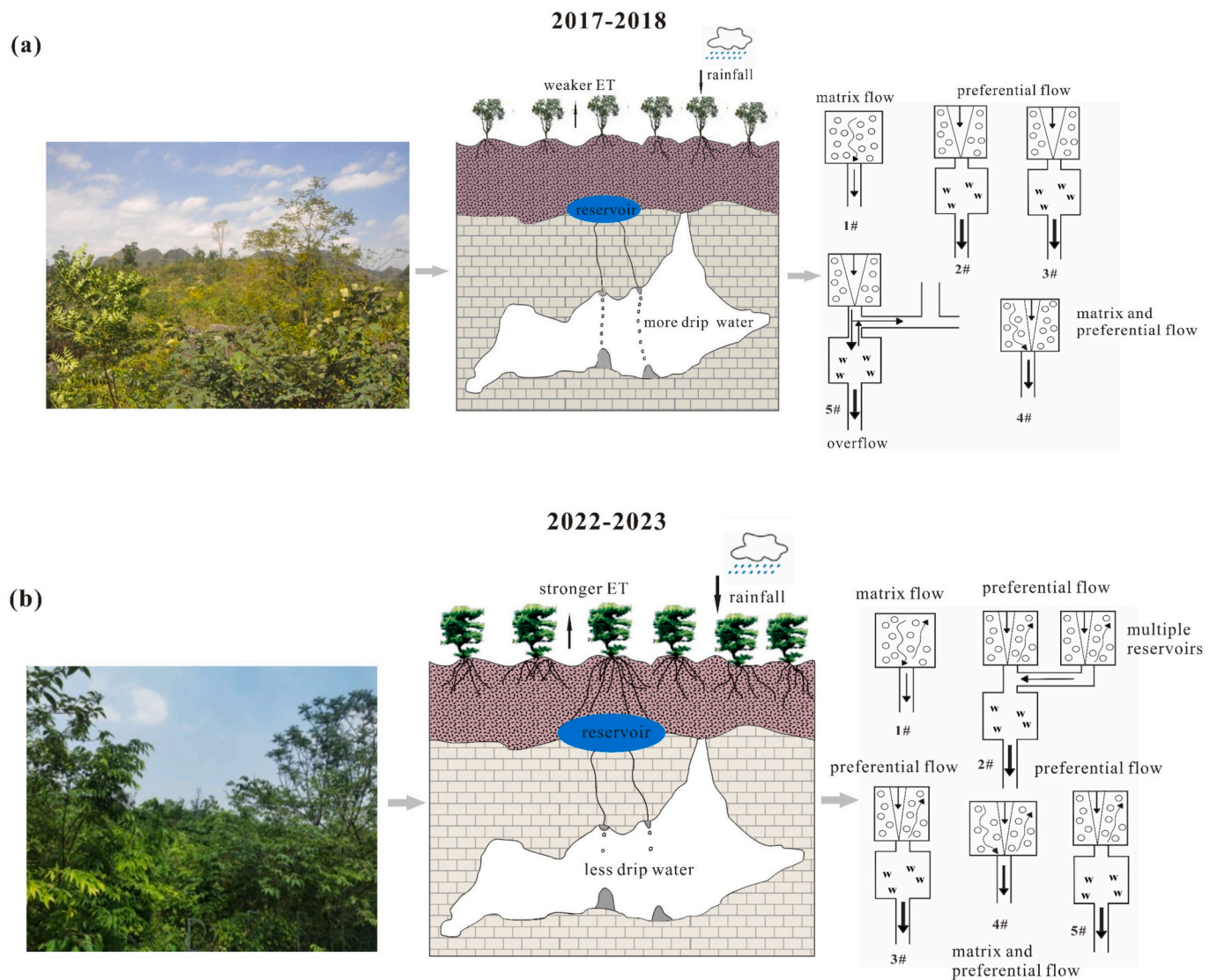


Fig. 9. Comparison of the hydrological conceptual model of Shawan Cave according to [Tooth and Fairchild \(2003\)](#). (a) Hydrological conceptual model when overlying vegetation was less. (b) Hydrological conceptual model when overlying vegetation was more. The vegetation in 2017–2018 was low scrub, whereas by 2022–2023 it had developed into approximate mature scrub and tree. The arrows in matrix and preferential flows represent evapotranspiration (ET).

changes may be affected by the seasonal change in drip water that is a function of water movement and water reservoirs ([Bradley et al., 2010](#)). Thus, seasonal changes in drip water, particularly in water movement and reservoir dynamics, are important factors controlling the variability of drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element/Ca values. To better elaborate on the key factors influencing drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element/Ca values, we analyze them across three distinct phases spanning 2017–2023: the initial phase (2017–2020), the later phase (2021–2023), and the long-term period (2017–2023).

During the initial phase (2017–2020), regardless of rainfall intensity, all drip sites except 1# retained water to some extent. Sufficient drip water was collected to document variations in drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca values from June 2017 to December 2018, implying that no interruption of drip water in Shawan Cave was observed during the dry season. This may be because less vegetation had shallow root system, and thus less water in reservoirs was absorbed, whereas most water infiltrated into the cave to form drip water ([Baker et al., 2021](#)). In previous studies, we found that drip water $\delta^{13}\text{C}_{\text{DIC}}$ was primarily controlled by cave air $p\text{CO}_2$ and the oscillation in drip water Mg/Ca and Sr/Ca were controlled by inverse-phase seasonal changes between rainfall amount and cave air $p\text{CO}_2$ ([Lyu et al., 2020](#), [Lyu et al., 2023](#)). We

introduce the line conditional excess (lc-excess), which can infer evaporation fractionation in different water samples. The lc-excess is calculated based on the formula ($\text{lc-excess} = \delta^2\text{H} - (a\delta^{18}\text{O} + b)$, where a and b are the slope and intercept of the local meteoric water line, respectively) ([Landwehr and Coplen, 2004](#)). Lower lc-excess values indicate a greater degree of evaporation during hydrological processes ([Hu et al., 2025](#)). [Fig. 10b](#) shows the time series of lc-excess values in drip water. The lc-excess values were lower and more discrete during 2017–2020. These results indicate that drip water was affected by evaporation and adequate mixing of rainwater from different rainfall intensities during 2017–2020. When strong mixing occurs in the aquifer, evaporation enriches seepage water in heavier isotopes compared with rainfall. Consequently, drip water $\delta^{18}\text{O}$ could not completely reflect changes in rainfall $\delta^{18}\text{O}$ due to evaporation and extensive mixing effects ([Duan et al., 2016](#)). The d-excess values of drip water during 2017–2020 were lower ([Fig. 10a](#)), which further suggests that drip water was affected by strong evaporation ([Chen et al. 2023](#); [Sun et al. 2018](#)).

During the later phase (2021–2023), drip water flow became intermittent. Consequently, insufficient water was collected to document the variations in drip water drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca values. From June 2022 to December 2022, considering water

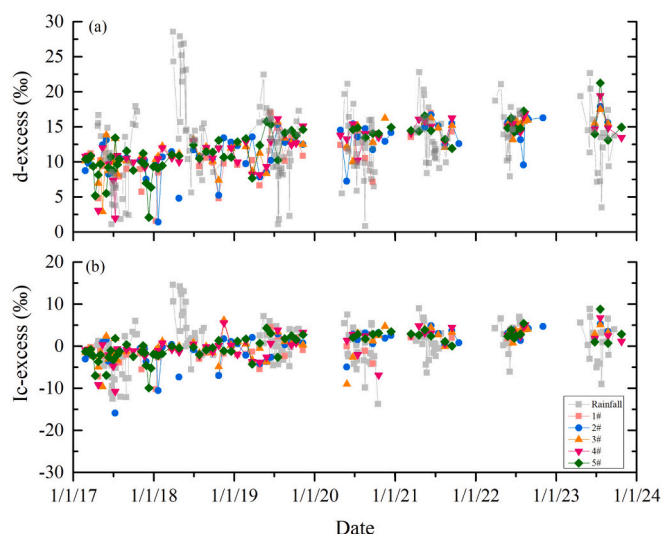


Fig. 10. Time series of d-excess values (a) and lc-excess values in rainfall and drip water (b) at Shawan Cave. Lower d-excess and lc-excess values indicate strong evaporation.

movement and reservoir evolution under different rainfall intensities, only site 2# could retain water regardless of rainfall intensity. Other drip sites lacked the capacity to store water and thus responded to rainfall only during heavy rain events. This may be attributed to the flourishing of vegetation, which strengthens and expands plant root systems, reducing soil moisture content and requiring greater rainfall for water to reach karst reservoirs and form cave drip water (Baker et al., 2021, Coleborn et al., 2016). As the water at 1#, 3#, 4#, and 5# drip sites is supplied by light and moderate rainfall events, the vegetation flourishing strengthens and expands plant root systems, resulting in the loss of infiltration water in the soil and vadose zones. During heavy rainfall events, activation of bypass preferential flow causes effective replenishment to feed drip water (Pape et al., 2010, Wang et al., 2018, Tian et al., 2024). However, the reservoir feeding the 2# site may evolve from a single reservoir to multiple reservoirs, resulting in interruption at site 2# occurring later than at other sites. The lc-excess values in drip water were higher and more convergent during 2021–2023 (Fig. 10b) and the d-excess values of drip water during 2021–2023 were higher (Fig. 10a), which implies lower evaporation as vegetation flourishes and weaker mixing effect of rainfall. Because mixing with antecedent water and evaporation had minimal impact, the occurrence of bypass preferential flow resulted in effective replenishment to feed drip water, which could occur only during heavy rainfall events (Tian et al., 2024). This is confirmed by Fig. 2d, which shows interruption of drip water in the dry season. Moreover, during June 2022–December 2022, the change in drip water $\delta^{18}\text{O}$ agreed with that of rainfall $\delta^{18}\text{O}$, further suggesting that drip water $\delta^{18}\text{O}$ was less affected by evaporation and mixing processes and thus can completely inherit the signal of rainfall $\delta^{18}\text{O}$ from heavy rainfall in the wet season. As previously mentioned, the changes in drip water $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca in the later phase were consistent with those in the initial phase.

During the long-term period (2017–2023), the $\delta^{18}\text{O}$ values of drip water shifted from a narrow to a broader range. Concurrently, both lc-excess and d-excess values exhibited an increasing trend (Fig. 10). This pattern suggests a decrease in the degree of evapotranspiration as vegetation gradually improved. However, the lc-excess and d-excess values shifted from being more discrete to more convergent, implying a weakening of the drip water mixing effect over time. The drip water $\delta^{13}\text{C}_{\text{DIC}}$ values became progressively more positive over the long-term period, a trend attributed to lower cave air $p\text{CO}_2$ driven by ventilation. The cave air $p\text{CO}_2$ reached its lowest levels during June–August each year and showed highest levels during October–November each

year, a pattern that is supported by the wind speed (Fig. S5 in Supplementary Material). From 2017 to 2021, its mean values were 1758, 2911, 2529, 1577, and 1201 ppm, respectively, showing a decreasing trend. Correspondingly, the drip water $\delta^{13}\text{C}_{\text{DIC}}$ values became progressively more positive from 2017 to 2023. The drip water Mg/Ca and Sr/Ca values were higher between 2017 and 2020, but lower between 2021 and 2023. This trend is consistent with the change in rainfall amount, which increased during the first period (2017–2020) and decreased during the second (2021–2023).

In short, a comparison of drip rates and proxy data between the 2017–2020 and 2021–2023 periods reveals that seasonal interruptions occurred at all sites in Shawan Cave, attributed to changes in reservoir. The evolution of the reservoirs—from water-retaining to water-limited under increased vegetation cover—resulted in drip water interruptions. Furthermore, during 2017–2020, the $\delta^{18}\text{O}$ of drip water at the five sites did not fully reflect rainfall $\delta^{18}\text{O}$ variations due to strong evaporation and mixing of rainwater from different rainfall intensities. In contrast, during 2021–2023, the five drip sites responded to heavy rainfall events, likely due to deeper root systems promoting preferential flow. These changes were mainly influenced by vegetation density, suggesting a shift from annual-scale averaging toward event-scale sensitivity in drip water response. Finally, integrated weekly and long-term data reveal that vegetation development led to a weakened mixing effect and the onset of seasonal interruptions.

5.3. The groundwater $\delta^{18}\text{O}$ relationship between Shawan Cave and Shawan Test site

Previous studies conducted at the Shawan Test Site have primarily emphasized the influence of surface land cover on groundwater $\delta^{18}\text{O}$ under controlled experimental conditions. The absence of soil and vegetation in bare rock land facilitated rapid groundwater recharge, resulting in minimal evaporation and, consequently, the most depleted groundwater $\delta^{18}\text{O}$ values. In contrast, land covers from bare rock to densely vegetation cover show higher evapotranspiration, leading to a corresponding enrichment of groundwater $\delta^{18}\text{O}$ (Hu et al., 2025). This controlled experiment only demonstrates the effect of vegetation cover on groundwater $\delta^{18}\text{O}$ on a spatial scale. However, a seven-year dataset of drip water $\delta^{18}\text{O}$ from Shawan Cave provides a temporal perspective. From 2017 to 2022, the annual mean $\delta^{18}\text{O}$ values in drip water were -9.25‰ , -9.01‰ , -8.58‰ , -8.58‰ , -9.11‰ , and -8.42‰ . This progressive enrichment coincided with vegetation recovery but did not correlate with variations in the annual mean $\delta^{18}\text{O}$ of rainfall (Table S1 in Supplementary Material). These results demonstrate that vegetation recovery can modulate drip water flow rate and paths and thereby drive the enrichment of drip water $\delta^{18}\text{O}$. These findings are consistent with the results of groundwater $\delta^{18}\text{O}$ in Shawan Test Site. However, the drip water $\delta^{18}\text{O}$ in Shawan Cave during 2017–2020 showed relatively smoothed variations, reflecting an integration of seasonal rainfall inputs (Fig. 3). The drip water $\delta^{18}\text{O}$ during 2021–2023 increasingly aligned with the isotopic composition of heavy summer monsoon rainfall (Figs. 6–8), suggesting a shift from annual-scale averaging toward event-scale sensitivity. These changes are different from groundwater $\delta^{18}\text{O}$ in Shawan Test Site, which groundwater $\delta^{18}\text{O}$ in bare karst could be the conservation of rainfall $\delta^{18}\text{O}$ seasonality. This may be because the simulation setup cannot fully capture the complexity and heterogeneity of natural karst cave, which leads to marked differences in $\delta^{18}\text{O}$ change between Shawan Cave and Shawan Test Site. Therefore, the variations in groundwater $\delta^{18}\text{O}$ observed at both Shawan Cave and the Shawan Test Site indicate that the less vegetation cover are more conducive to rapid drip water infiltration, but the preservation of seasonal rainfall $\delta^{18}\text{O}$ signals were different.

However, the findings of Hu et al. (2025), which are derived from artificial shallow settings and the natural conditions of Shawan Cave, may not be directly applicable to deeper cave systems such as Furong Cave described by Li et al. (2011), where the upper vadose zone is

300–500 m thick. This is because the complete mixing of water from different rainfall events in the aquifer leads to the lack of a seasonal signal in drip water $\delta^{18}\text{O}$. In such deep systems, evaporation driven by vegetation cover is not the main factor influencing the drip water $\delta^{18}\text{O}$. Consequently, ignoring the thickness of the cave ceiling risks misinterpreting speleothem proxies. This is particularly true in monsoon-affected regions where rainfall is highly seasonal, and infiltration is strongly influenced by vegetation cover. The thickness of the cave ceiling adds a crucial dimension to the interpretation of speleothem proxies. A thin ceiling, combined with the recovery of vegetation and soil moisture thresholds, can lead to drip water dynamics that interrupt discharge.

5.4. Implications of drip water dynamics and geochemical proxies on speleothem records

Drip water is the material basis for the formation of speleothems, as speleothems are calcite deposited from drip water (Meyer et al., 2014). Revealing the seasonality of drip water discharge is essential to understand the mechanisms of speleothem formation. Our results show that drip water in Shawan Cave was gradually interrupted from 2017 to 2023 and evolved to occur in the wet season with little in the dry season. These observations are ascribed to deeper root systems activating bypass preferential flows as vegetation gradually flourished, rather than to changes in annual rainfall amount. This is primarily because the thin roof of Shawan Cave allows well-developed root systems. The amount of calcite deposition displayed obvious seasonal variation in Shawan Cave: it was higher from June to August, lower from September to October, and minimal or absent from November to May of the following year (Fig. S4 in Supplementary Material). We can infer the stalagmites from Shawan Cave can be expected to show strong seasonality with continuous growth in the wet season and occurring hiatuses in the dry season (Miller et al., 2021). Based on multi-year dry-season rainfall and drip water behavior data from Shawan Cave, we found that dry-season rainfall was low during 2018–2019, yet drip water remained sufficient in that period. However, during 2022–2023, dry-season rainfall was also low, and drip water was interrupted in that period (Fig. S2 in Supplementary Material). These results suggest that stalagmite hiatuses are more likely caused by the extent of vegetation flourishing rather than by changes in annual rainfall amount. The transition from cultivated crops to dense shrubs around Shawan Cave results in young vegetation with potentially higher water demand than a mature forest. This elevated demand can cause drip water interruption and stalagmite hiatuses specifically during this transitional period. Such hiatuses likely correspond to vegetation transition phases, particularly where the cave roof is thin. Therefore, we propose using the thickness of the overlying bedrock as a discerning criterion for selecting the caves to reconstruct paleoclimate. When the overlying bedrock is sufficiently thick to allow adequate infiltration, continuous annual speleothem deposition can occur. In contrast, if the bedrock is too thin, causing interruption of drip water in the dry season, speleothem growth will cease when drips become inactive.

Karst hydrological discharge pattern and behavior allow us to understand the sensitivity of drip water and the evolution of the reservoirs feeding drip water to different rainfall intensities (Baldini et al., 2006). The prerequisite for applying speleothem proxies to reconstruct paleoclimate changes is the response of drip water proxies to signals of outside rainfall (Fairchild et al., 2006). Consequently, drip water discharge patterns under different rainfall intensities could be used to assess the extent to which drip water proxies inherit the rainfall signal and directly affect the interpretation of speleothem proxies. Hydrological drip rate data from Shawan Cave allow us to deduce water movement and reservoir evolution associated with individual drip sites, indicating that different water pathways and mixing proportions between new and old water in reservoirs lead to different magnitudes of drip water proxies. For example, based on 11 years of monitoring in Yuhua Cave, Southeast

China, it was found that drip water $\delta^{18}\text{O}$ showed non-stationary behavior in residence time and reservoir ratios. This variation may be regulated by significant changes in local rainfall, with increased rainfall leading to greater amplitude of drip water $\delta^{18}\text{O}$ (Qiu et al., 2023). In addition, our study revealed that drip water $\delta^{18}\text{O}$ in heavy rainfall of the wet season could completely inherit the signal of rainfall $\delta^{18}\text{O}$. Therefore, speleothem $\delta^{18}\text{O}$ in Shawan Cave could record the changes in the summer monsoon, as increased rainfall related to strong monsoon changes can be reflected in speleothem $\delta^{18}\text{O}$ (Yin et al., 2023). In contrast, speleothem $\delta^{18}\text{O}$ in Shawan Cave in 2018 recorded an entire year or more of rainfall accumulation and can be regarded as a proxy for annual or multi-year rainfall amount due to evaporation and adequate mixing of rainwater from different rainfall intensities.

For each drip water site, the drip rate at the 2# site during light, moderate, and heavy rainfall in 2022 was higher than that of 2018. This may be attributed to the reservoir feeding the 2# drip site having evolved into multiple reservoirs. That is, the aquifer storage may not remain constant over time, implying that the speleothem proxies growing beneath the drip site may evolve to reflect seasonal climate signals in the wet season. Accounting for the evolution of water reservoirs could therefore help explain the shifts in speleothem proxies. The drip rate at the 3#, 4#, and 5# sites during light and moderate in 2018 is higher than that of 2022, which may be associated with vegetation roots absorbing the infiltration water. This implies that small rainfall events may be lost to evapotranspiration before infiltrating into the cave, leading to more positive drip water $\delta^{18}\text{O}$ over time. As vegetation improved, evapotranspiration increased, which reduced direct evaporation from the soil and enhanced the contribution of isotopically heavier rainfall to infiltration. The speleothem $\delta^{18}\text{O}$ fed by these drip sites may not fully record the signal of light and moderate rainfall $\delta^{18}\text{O}$. However, the drip rates at the 4# and 5# sites during heavy rainfall were relatively high due to deeper root systems driving the occurrence of preferential flow, suggesting that the drip water proxies from these sites may reflect heavy rainfall events, particularly severe flooding events. At present, the thin overlying bedrock in Shawan Cave makes the 4# and 5# drip sites optimal locations to observe speleothem proxies for reconstructing severe rainfall events, due to deeper root systems driving preferential flow as vegetation gradually flourishes. However, the 2# drip site is an optimal location for a speleothem proxy to record seasonal and annual changes in rainfall.

Interpreting the signals in speleothem proxies in different caves and different stalagmite locations likely requires a comparative analysis of different periods and locations of drip water with different drip rates. More importantly, it is advisable to recognize that discontinuities in stalagmite growth may be ascribed to the seasonality of drip water flow, which is controlled by variations in karst residence reservoirs or shifts in water pathways. Therefore, we emphasize the necessity of understanding drip rate patterns under different rainfall intensities and establishing the relationship between drip rate variability and drip water $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, and element/Ca in high-resolution speleothem reconstructions to derive precise estimates of hydrological conditions. Nevertheless, this study is constrained by limited high-frequency sampling intervals, the absence of direct soil and vegetation monitoring, and the focus on a single shallow cave, which should be considered when extrapolating the findings to broader paleoclimate reconstructions.

6. Conclusions

This study combined high-resolution monitoring of drip rates, $\delta^{18}\text{O}$, $\delta^{13}\text{C}_{\text{DIC}}$, Mg/Ca, and Sr/Ca of drip water to investigate how rainfall and vegetation changes shape drip water dynamics in Shawan Cave between 2017 and 2023. From 2017 to 2019, drip water was present throughout both wet and dry seasons, but from 2020 to 2023 flow became intermittent, occurring mainly during the wet season and largely absent in the dry season. We attribute this shift to vegetation recovery, where deeper root systems altered soil moisture thresholds and increased the

rainfall required for infiltration. These changes reshaped the reservoirs feeding drip water: the 2# site evolved from a single reservoir to multiple sources, while other sites responded only to heavy rainfall as shallow infiltration was increasingly intercepted by roots and preferential flow. Our findings demonstrate that vegetation recovery, rather than rainfall amount alone, is the dominant control on seasonal drip water interruption in this shallow cave. Vegetation recovery, by influencing evapotranspiration, can modulate drip water flow paths and thereby drive the enrichment of drip water $\delta^{18}\text{O}$. In addition, drip water $\delta^{18}\text{O}$ records evolved over time: from 2017 to 2018 they reflected integrated annual rainfall influenced by evaporation and mixing, whereas by 2022 they more directly recorded heavy summer monsoon rainfall. This progression illustrates how non-stationary infiltration pathways can shift speleothem $\delta^{18}\text{O}$ from recording long-term rainfall averages to event-scale signals.

Our findings reveal the key regulatory role of vegetation restoration on karst vadose zone hydrology, showing that it interrupts seasonal drip water flow by altering infiltration pathways and aquifer water storage. This work serves as an empirical case study connecting non-stationary reservoir change driven by hydrological processes with vegetation recovery. Our findings thereby enhance the framework for cave monitoring and paleoclimate reconstruction in monsoon regions. A limitation is that the overlying vegetation at Shawan Cave is transitional, so the applicability of these findings to caves with other mature vegetation remains to be tested.

CRedit authorship contribution statement

Yina Lyu: Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Weijun Luo:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Reni Ustiatik:** Writing – review & editing, Visualization, Formal analysis. **Guangjie Luo:** Writing – review & editing, Resources, Investigation. **Alena Gessert:** Writing – review & editing, Visualization. **Guangneng Zeng:** Visualization, Methodology. **Yanwei Wang:** Methodology, Investigation. **Jia Chen:** Visualization, Data curation. **Anyun Cheng:** Writing – review & editing, Resources. **Shijie Wang:** Visualization, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2026.135322>.

Data availability

Data will be made available on request.

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